

BETHLEHEM, South AfricaWMO: **684610**Lat: **28.2497S**Lon: **28.3342E**Elev: **5535**StdP: **11.98**Time Zone: **2.00 (E02)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	25.2	27.8	8.4	10.4	52.5	12.2	12.5	49.3	21.6	51.8	19.0	51.8	2.1	50	0.465

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	21.1	85.5	59.2	83.3	58.9	81.2	59.0	65.7	75.6	64.7	74.7	63.8	73.6	9.3	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
62.9	105.7	68.3	61.9	101.9	67.5	60.9	98.3	66.8	33.9	75.9	33.0	74.7	32.2	73.8	70.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 19.7	17.1	15.3	21.1	89.7	2.9	3.0	19.0	91.9	17.4	93.7	15.7	95.4	13.6	97.6		
(5)			19.1	67.8	2.7	1.6	17.2	68.9	15.6	69.9	14.1	70.8	12.1	71.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	58.1	67.4	66.8	63.9	57.6	51.6	46.1	45.5	50.6	57.2	61.2	63.6	66.2	
	DBStd	8.90	3.79	3.39	3.81	4.53	4.67	3.88	4.26	5.31	5.42	5.17	5.15	4.30	
	HDD50	380	0	0	0	4	37	126	147	57	7	1	1	0	
	HDD65	2858	20	18	64	224	416	567	605	448	238	138	84	36	
	CDD50	3332	540	470	432	231	86	10	7	74	224	347	408	503	
	CDD65	334	94	68	31	2	0	0	0	0	5	19	41	74	
	CDH74	3868	743	531	350	69	8	0	0	27	297	497	600	746	
	CDH80	714	152	86	38	2	0	0	0	0	34	98	131	173	
Wind	WSAvg	6.6	7.3	6.7	6.3	5.5	5.3	5.3	5.6	6.3	7.2	7.8	7.9	7.5	
Precipitation	PrecAvg	27.2	4.8	3.9	3.0	1.6	0.7	0.4	0.3	0.7	1.1	3.0	3.7	4.2	
	PrecMax	39.6	9.6	9.4	8.3	4.6	4.6	2.2	2.3	4.6	7.5	8.8	9.7	10.6	
	PrecMin	17.2	1.1	0.3	0.8	0.0	0.0	0.0	0.0	0.0	0.0	0.6	0.4	0.9	
	PrecStd	5.2	2.3	2.0	1.6	1.1	0.9	0.6	0.5	1.1	1.3	2.0	2.1	1.9	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	87.9	86.1	84.1	79.2	74.9	68.9	69.1	77.4	83.6	86.0	87.1	89.8	
		MCWB	60.9	60.4	60.3	56.1	51.5	47.1	46.5	51.4	53.9	56.2	57.7	59.7	
	2%	DB	84.0	82.7	80.6	76.0	71.0	65.5	65.9	73.3	80.6	82.8	83.8	84.3	
		MCWB	61.5	61.2	60.1	56.0	51.2	46.4	45.7	50.1	53.2	55.4	57.8	59.9	
	5%	DB	81.0	80.0	78.0	73.4	68.5	62.9	63.5	69.9	77.7	79.7	80.6	81.1	
		MCWB	62.0	61.7	59.9	56.1	50.4	45.6	45.1	48.4	52.4	55.3	57.8	60.0	
	10%	DB	78.5	77.4	75.4	70.6	65.9	60.7	60.9	66.8	74.4	76.4	77.4	78.3	
		MCWB	62.2	61.8	60.1	55.5	49.7	44.9	44.4	47.4	51.3	54.9	57.8	60.5	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	67.3	67.1	65.5	61.2	56.5	51.1	50.1	55.1	59.2	62.0	64.6	66.4
			MCDB	77.5	75.9	74.4	70.7	66.5	61.8	61.1	69.3	75.5	73.7	75.5	77.4
2%		WB	65.7	65.5	63.8	59.7	54.3	49.3	48.2	52.8	56.6	60.2	62.7	64.6	
		MCDB	76.1	74.8	73.6	69.6	64.9	60.1	60.6	66.9	72.3	72.6	73.8	75.2	
5%		WB	64.7	64.4	62.6	58.5	52.7	47.7	46.8	51.0	54.7	58.9	61.5	63.4	
		MCDB	74.9	73.9	72.0	68.5	63.7	58.8	59.9	64.7	71.0	71.3	72.5	73.9	
10%		WB	63.7	63.4	61.6	57.3	51.2	46.2	45.3	49.4	53.2	57.5	60.3	62.2	
		MCDB	73.4	72.4	70.8	67.1	62.4	57.7	58.6	62.5	69.1	69.8	71.1	72.7	
Mean Daily Temperature Range	5% DB	MDBR	21.1	20.8	21.1	23.8	27.4	28.7	29.5	30.1	30.1	26.2	24.1	22.4	
		MCDBR	26.2	26.1	26.6	29.1	32.0	32.6	33.7	34.7	35.2	33.1	30.0	27.6	
		MCWBR	9.1	9.6	10.1	12.3	15.2	16.9	17.3	16.2	14.8	12.3	10.1	9.7	
	5% WB	MCDBR	21.8	20.9	21.4	22.4	26.1	28.0	29.2	29.5	29.7	25.1	23.6	22.7	
		MCWBR	8.9	8.9	9.5	10.7	13.7	16.1	16.1	15.1	13.5	10.9	9.7	9.6	
Clear-Sky Solar Irradiance	taub	0.359	0.355	0.342	0.304	0.273	0.257	0.257	0.319	0.358	0.388	0.345	0.350		
	taud	2.380	2.395	2.403	2.486	2.555	2.565	2.544	2.358	2.254	2.215	2.380	2.396		
	Ebn at Noon	312	307	301	298	296	294	299	289	291	294	314	315		
	Edh at Noon	41	39	37	31	26	24	25	34	42	47	41	41		
All-Sky Solar Radiation	RadAvg	2164	2015	1771	1486	1304	1179	1287	1521	1855	2008	2215	2236		
	RadStd	205	184	130	109	67	63	52	84	96	150	145	141		

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	+1.03	+1.28	N/A	N/A	-0.70	-1.01	N/A	N/A	N/A
(47)	Regional (6 neighbors)	+1.03	+1.28	N/A	N/A	-0.70	-1.01	N/A	N/A	N/A

Nomenclature: See separate page